

Max Callaghan

Hermannstrasse 57

12049 Berlin

📞 +49 01575 032 0056

✉ max.w.callaghan@gmail.com

Education

- 2018–2022 **PhD**, *University of Leeds*, School of Earth and Environment
Machine Reading the Science of Climate Change: Computational Tools to Support Evidence-Based Decision-Making in the Age of Big Literature
Supervisors: Professor Jan Minx, Professor Piers Forster
- 2013–2016 **Master of Public Policy**, *Hertie School of Governance*, Berlin, 91/100
- 2007–2011 **BA - English and Related Literature**, *University of York*, First-class honours

Employment History

- 2025– **Scientist**, *Potsdam Institute for Climate Impacts Research*, Potsdam
- 2016–2024 **Scientist**, *Mercator Research on Global Commons and Climate Change (MCC)*, Berlin
- 2015–2016 **Research Assistant**, *Mercator Research on Global Commons and Climate Change (MCC)*, Berlin
- 2016–2016 **Teaching Assistant – Stats II**, *Hertie School of Governance*, Berlin
- 2015–2015 **Teaching Assistant – Applied Economic Analysis**, *Hertie School of Governance*
- 2014–2015 **Junior Research Fellow – Development, Gender and Human Rights**, *United Nations System Staff College (UNSSC)*, Turin
- 2014–2014 **Research Assistant – The Governance Report**, *Hertie School of Governance*, Berlin
- 2013–2014 **Research Analyst**, *Market Logic Software*, Berlin
- 2013–2013 **Junior Consultant**, *Cadence Partnership*, London
- 2012–2013 **Communications Intern**, *Release*, London
- 2011–2012 **Teacher**, *The English School of Mongolia*, Ulaanbaator

Honors and Scholarships

- 2017 PhD Scholarship, Heinrich Böll Stiftung
- 2016 Dean's List, Hertie School of Governance

Funding

- 2024–2026 **DESTINY**, *Co-applicant and work package lead*, £10,000,000 - funded by the Wellcome Trust
- 2024–2026 **Pathfinder 2**, *Co-applicant*, £3,500,000 - funded by the Wellcome Trust

- 2023–2024 **Digital Evidence Synthesis Tools in Climate and Health**, *Co-applicant*, £320,000
- funded by the Wellcome Trust
- 2021–2022 **Climate Vulnerability Monitor**, *Design and lead of Work Package 1 - Observed climate change*, €669,390 - funded by Climate Vulnerable Forum

Teaching Experience

- Autumn 2023 **Methods in Planetary Health**, *Planetary Health MsC*, London School of Hygiene and Tropical Medicine, London, Guest Lecturer (1 Session)
- Autumn 2023 **Text as Data**, *Data Science for Public Policy MsC*, Hertie School of Governance, Berlin, Guest Lecturer (12 Sessions)
- Autumn 2022 **Text as Data**, *Data Science for Public Policy MsC*, Hertie School of Governance, Berlin, Guest Lecturer (12 Sessions)
- Summer 2022 **NLP for evidence synthesis in climate change research**, *Climate Change AI Summer School*, Online, Tutorial

Mentoring Experience

- 2024 **Co-supervision of Master Thesis**, *Pre-train, prompt and predict: leveraging stochastic parrots for multi-label document classification*, Santiago Sordo Ruz, Hertie School of Governance
- 2021 **Co-supervision of Master Thesis**, *The dynamics of hate speech against Greta Thunberg: An analysis of Tweets*, Nina Locher, Hertie School of Governance

Research Service

- Editorial I am a Methods Editor (with speciality in AI) for Campbell Systematic Reviews. I am also guest editing a special issue on AI methods for systematic reviews there.
- Peer Review I have provided reviews for Nature Climate Change, Nature Sustainability, Energy Research & Social Science, Environmental Research Letters, Climatic Change, Regional Environmental Change, Research Synthesis Methods, The Journal of Hydrology, and the Journal of the American Medical Informatic Association
- Assessments I was a contributing author in Chapter 9 of Working Group II's contribution to the sixth assessment report of the Intergovernmental Panel on Climate Change. I was a contributing author in Chapter 3 of the Austrian Panel on Climate Change's special report "Strukturen für ein klimafreundliches Leben"

Media Echo

My work has been covered by the Washington Post, CBC, CNN, Al Jazeera, ABC News, and the Frankfurter Allgemeine Zeitung.

Peer Reviewed Publications

Up-to-date publications and statistics are available at <https://scholar.google.com/citations?user=B6mNFrAAAAAJ&hl=en>).

- [1] Sarah Lück, Max Callaghan, Carl-Friedrich Schleussner, and Jan Minx. The evidence gap index:

- Mapping evidence where it matters for climate change impacts. *Climate and Development*, 0(0):1–9, 2025. ISSN 1756-5529. doi: 10.1080/17565529.2025.2479005.
- [2] Simon Montfort, **Max Callaghan**, Felix Creutzig, William F. Lamb, Chenxi Lu, Tim Repke, Ke Ge, and Jan Minx. Systematic global stocktake of over 50,000 urban climate change studies. *Nature Cities*, 2(7):613–625, July 2025. ISSN 2731-9997. doi: 10.1038/s44284-025-00260-8.
- [3] James D. Ford, Robbert Biesbroek, Lea Berrang Ford, Felix Creutzig, Neal Haddaway, Sherilee Harper, Jan C. Minx, Mark New, Anne J. Sietsma, Carol Zavaleta-Cortijo, and **Max Callaghan**. Recommendations for producing knowledge syntheses to inform climate change assessments. *Nature Climate Change*, 15(7):698–708, July 2025. ISSN 1758-6798. doi: 10.1038/s41558-025-02354-6.
- [4] **Max Callaghan**, Lucy Banisch, Niklas Doebbeling-Hildebrandt, Duncan Edmondson, Christian Flachslund, William F. Lamb, Sebastian Levi, Finn Müller-Hansen, Eduardo Posada, Shraddha Vasudevan, and Jan C. Minx. Machine learning map of climate policy literature reveals disparities between scientific attention, policy density, and emissions. *npj Climate Action*, 4(1): 1–14, February 2025. ISSN 2731-9814. doi: 10.1038/s44168-024-00196-0.
- [5] Sarah Lück, **Max Callaghan**, Malgorzata Borchers, Annette Cowie, Sabine Fuss, Matthew Gidden, Jens Hartmann, Claudia Kammann, David P. Keller, Florian Kraxner, William F. Lamb, Niall Mac Dowell, Finn Müller-Hansen, Gregory F. Nemet, Benedict S. Probst, Phil Renforth, Tim Repke, Wilfried Rickels, Ingrid Schulte, Pete Smith, Stephen M. Smith, Daniela Thrän, Tiffany G. Troxler, Volker Sick, Mijndert van der Spek, and Jan C. Minx. Scientific literature on carbon dioxide removal revealed as much larger through AI-enhanced systematic mapping. *Nature Communications*, 16(1):6632, July 2025. ISSN 2041-1723. doi: 10.1038/s41467-025-61485-8.
- [6] Sarah Young, Malgorzata Lagisz, **Max W. Callaghan**, Shinichi Nakagawa, and Neal R. Haddaway. Collaboration and term usage dynamics in agricultural buffer strip research: Research weaving. *Ecological Solutions and Evidence*, 5(3):e12376, 2024. ISSN 2688-8319. doi: 10.1002/2688-8319.12376.
- [7] Kim R. van Daalen, Cathryn Tonne, Jan C. Semenza, Joacim Rocklöv, Anil Markandya, Niheer Dasandi, Slava Jankin, Hicham Achebak, Joan Ballester, Hannah Bechara, Thessa M. Beck, **Max W. Callaghan**, Bruno M. Carvalho, Jonathan Chambers, Marta Cirah Pradas, Orin Courtenay, Shouro Dasgupta, Matthew J. Eckelman, Zia Farooq, Peter Fransson, Elisa Gallo, Olga Gasparyan, Nube Gonzalez-Reviriego, Ian Hamilton, Risto Hänninen, Charles Hatfield, Kehan He, Aleksandra Kazmierczak, Vladimir Kendrovski, Harry Kennard, Gregor Kiesewetter, Rostislav Kouznetsov, Hedi Katre Kriit, Alba Llabrés-Brustenga, Simon J. Lloyd, Martín Lotto Batista, Carla Maia, Jaime Martinez-Urtaza, Zhifu Mi, Carles Milà, Jan C. Minx, Mark Nieuwenhuijsen, Julia Palamarchuk, Dafni Kalatzi Pantera, Marcos Quijal-Zamorano, Peter Rafaj, Elizabeth J. Z. Robinson, Nacho Sánchez-Valdivia, Daniel Scamman, Oliver Schmoll, Maquins Odhiambo Sewe, Jodi D. Sherman, Pratik Singh, Elena Sirotkina, Henrik Sjödin, Mikhail Sofiev, Balakrishnan Solaraju-Murali, Marco Springmann, Marina Treskova, Joaquin Triñanes, Eline Vanuytrecht, Fabian Wagner, Maria Walawender, Laura Warnecke, Ran Zhang, Marina Romanello, Josep M. Antó, Maria Nilsson, and Rachel Lowe. The 2024 Europe report of the Lancet Countdown on health and climate change: Unprecedented warming demands unprecedented action. *The Lancet Public Health*, 9(7):e495–e522, July 2024. ISSN 2468-2667. doi: 10.1016/S2468-2667(24)00055-0.

- [8] **Max Callaghan**, Finn Müller-Hansen, Melissa Bond, Candyce Hamel, Declan Devane, Wojciech Kusa, Alison O'Mara-Eves, Rene Spijker, Mark Stevenson, Claire Stansfield, James Thomas, and Jan C. Minx. Computer-assisted screening in systematic evidence synthesis requires robust and well-evaluated stopping criteria. *Systematic Reviews*, 13(1):284, November 2024. ISSN 2046-4053. doi: 10.1186/s13643-024-02699-7.
- [9] Tim Repke, **Max Callaghan**, William F. Lamb, Sarah Lück, Finn Müller-Hansen, and Jan C. Minx. How global crises compete for our attention: Insights from 13.5 million tweets on climate change during COVID-19. *Energy Research & Social Science*, 116:103668, October 2024. ISSN 2214-6296. doi: 10.1016/j.erss.2024.103668.
- [10] Marina Romanello, Maria Walawender, Shih-Che Hsu, Annalyse Moskeland, Yasna Palmeiro-Silva, Daniel Scamman, Zakari Ali, Nadia Ameli, Denitsa Angelova, Sonja Ayeb-Karlsson, Sara Basart, Jessica Beagley, Paul J. Beggs, Luciana Blanco-Villafuerte, Wenjia Cai, **Max Callaghan**, Diarmid Campbell-Lendrum, Jonathan D. Chambers, Victoria Chicmana-Zapata, Lingzhi Chu, Troy J. Cross, Kim R. van Daalen, Carole Dalin, Niheer Dasandi, Shouro Dasgupta, Michael Davies, Robert Dubrow, Matthew J. Eckelman, James D. Ford, Chris Freyberg, Olga Gasparyan, Georgiana Gordon-Strachan, Michael Grubb, Samuel H. Gunther, Ian Hamilton, Yun Hang, Risto Hänninen, Stella Hartinger, Kehan He, Julian Heidecke, Jeremy J. Hess, Louis Jamart, Slava Jankin, Harshavardhan Jatkar, Ollie Jay, Ilan Kelman, Harry Kennard, Gregor Kiesewetter, Patrick Kinney, Dominic Kniveton, Rostislav Kouznetsov, Pete Lampard, Jason K. W. Lee, Bruno Lemke, Bo Li, Yang Liu, Zhao Liu, Alba Llabrés-Brustenga, Melissa Lott, Rachel Lowe, Jaime Martinez-Urtaza, Mark Maslin, Lucy McAllister, Celia McMichael, Zhifu Mi, James Milner, Kelton Minor, Jan Minx, Nahid Mohajeri, Natalie C. Momen, Maziar Moradi-Lakeh, Karyn Morrissey, Simon Munzert, Kris A. Murray, Nick Obradovich, Megan B. O'Hare, Camile Oliveira, Tadj Oreszczyn, Matthias Otto, Fereidoon Owfi, Olivia L. Pearman, Frank Pega, Andrew J. Perishing, Ana-Catarina Pinho-Gomes, Jamie Ponmattam, Mahnaz Rabbaniha, Jamie Rickman, Elizabeth Robinson, Joacim Rocklöv, David Rojas-Rueda, Renee N. Salas, Jan C. Semenza, Jodi D. Sherman, Joy Shumake-Guillemot, Pratik Singh, Henrik Sjödin, Jessica Slater, Mikhail Sofiev, Cecilia Sorensen, Marco Springmann, Zélie Stalhandske, Jennifer D. Stowell, Meisam Tabatabaei, Jonathon Taylor, Daniel Tong, Cathryn Tonne, Marina Treskova, Joaquin A. Trinanes, Andreas Uppstu, Fabian Wagner, Laura Warnecke, Hannah Whitcombe, Peng Xian, Carol Zavaleta-Cortijo, Chi Zhang, Ran Zhang, Shihui Zhang, Ying Zhang, Qiao Zhu, Peng Gong, Hugh Montgomery, and Anthony Costello. The 2024 report of the Lancet Countdown on health and climate change: Facing record-breaking threats from delayed action. *The Lancet*, 0(0), October 2024. ISSN 0140-6736, 1474-547X. doi: 10.1016/S0140-6736(24)01822-1.
- [11] Wenjia Cai, Chi Zhang, Shihui Zhang, Yuqi Bai, **Max Callaghan**, Nan Chang, Bin Chen, Huiqi Chen, Liangliang Cheng, Hancheng Dai, Weicheng Fan, Dabo Guan, Yixin Hu, Yifan Hu, Junyi Hua, Cunrui Huang, Hong Huang, Jianbin Huang, Xiaomeng Huang, John S. Ji, Qiaolei Jiang, Xiaopeng Jiang, Gregor Kiesewetter, Tiantian Li, Bo Li, Lu Liang, Borong Lin, Hualiang Lin, Huan Liu, Qiyong Liu, Zhao Liu, Zhu Liu, Yanxiang Liu, Shuhan Lou, Bo Lu, Chenxi Lu, Zhenyu Luo, Zhifu Mi, Yanqing Miao, Chao Ren, Marina Romanello, Jianxiang Shen, Jing Su, Rui Su, Yuze Sun, Xinlu Sun, Maria Walawender, Can Wang, Qing Wang, Qiong Wang, Laura Warnecke, Wangyu Wei, Xiaohui Wei, Sanmei Wen, Yang Xie, Hui Xiong, Bing Xu, Xiu Yang, Yuren Yang, Fanghong Yao, Le Yu, Wenhao Yu, Jiacan Yuan, Yiping Zeng, Jing Zhang, Rui Zhang, Shangchen Zhang, Shaohui Zhang, Mengzhen Zhao, Qi Zhao, Qiang Zhao, Dashan Zheng, Hao Zhou, Jingbo Zhou, Ziqiao Zhou, Yong Luo, and Peng Gong. The 2024 China report of

the Lancet Countdown on health and climate change: Launching a new low-carbon, healthy journey. *The Lancet Public Health*, 9(12):e1070–e1088, December 2024. ISSN 2468-2667. doi: 10.1016/S2468-2667(24)00241-X.

- [12] Diana Danilenko, Marina Andrijevic, Anne J. Sietsma, **Max Callaghan**, and Tarun Khanna. Women are under-represented in adaptation policy research and are more likely to emphasise justice topics. *Environmental Research: Climate*, 3(4):045010, August 2024. ISSN 2752-5295. doi: 10.1088/2752-5295/ad6f3b.
- [13] Niklas Döbbeling-Hildebrandt, Klaas Miersch, Tarun M. Khanna, Marion Bachelet, Stephan B. Bruns, **Max Callaghan**, Ottmar Edenhofer, Christian Flachsland, Piers M. Forster, Matthias Kalkuhl, Nicolas Koch, William F. Lamb, Nils Ohlendorf, Jan Christoph Steckel, and Jan C. Minx. Systematic review and meta-analysis of ex-post evaluations on the effectiveness of carbon pricing. *Nature Communications*, 15(1):4147, May 2024. ISSN 2041-1723. doi: 10.1038/s41467-024-48512-w.
- [14] Marina Romanello, Claudia di Napoli, Carole Green, Harry Kennard, Pete Lampard, Daniel Scamman, Maria Walawender, Zakari Ali, Nadia Ameli, Sonja Ayeb-Karlsson, Paul J. Beggs, Kristine Belesova, Lea Berrang Ford, Kathryn Bowen, Wenjia Cai, **Max Callaghan**, Diarmid Campbell-Lendrum, Jonathan Chambers, Troy J. Cross, Kim R. van Daalen, Carole Dalin, Niheer Dasandi, Shouro Dasgupta, Michael Davies, Paula Dominguez-Salas, Robert Dubrow, Kristie L. Ebi, Matthew Eckelman, Paul Ekins, Chris Freyberg, Olga Gasparyan, Georgiana Gordon-Strachan, Hilary Graham, Samuel H. Gunther, Ian Hamilton, Yun Hang, Risto Hänninen, Stella Hartinger, Kehan He, Julian Heidecke, Jeremy J. Hess, Shih-Che Hsu, Louis Jamart, Slava Jankin, Ollie Jay, Ilan Kelman, Gregor Kiesewetter, Patrick Kinney, Dominic Kniveton, Rostislav Kouznetsov, Francesca Larosa, Jason K. W. Lee, Bruno Lemke, Yang Liu, Zhao Liu, Melissa Lott, Martín Lotto Batista, Rachel Lowe, Maquins Odhiambo Sewe, Jaime Martinez-Urtaza, Mark Maslin, Lucy McAllister, Celia McMichael, Zhifu Mi, James Milner, Kelton Minor, Jan C. Minx, Nahid Mohajeri, Natalie C. Momen, Maziar Moradi-Lakeh, Karyn Morrissey, Simon Munzert, Kris A. Murray, Tara Neville, Maria Nilsson, Nick Obradovich, Megan B. O'Hare, Camile Oliveira, Tadj Oreszczyn, Matthias Otto, Fereidoon Owfi, Olivia Pearman, Frank Pega, Andrew Pershing, Mahnaz Rabbaniha, Jamie Rickman, Elizabeth J. Z. Robinson, Joacim Rocklöv, Renee N. Salas, Jan C. Semenza, Jodi D. Sherman, Joy Shumake-Guillemot, Grant Silbert, Mikhail Sofiev, Marco Springmann, Jennifer D. Stowell, Meisam Tabatabaei, Jonathon Taylor, Ross Thompson, Cathryn Tonne, Marina Treskova, Joaquin A. Trinanes, Fabian Wagner, Laura Warnecke, Hannah Whitcombe, Matthew Winning, Arthur Wyns, Marisol Yglesias-González, Shihui Zhang, Ying Zhang, Qiao Zhu, Peng Gong, Hugh Montgomery, and Anthony Costello. The 2023 report of the Lancet Countdown on health and climate change: The imperative for a health-centred response in a world facing irreversible harms. *The Lancet*, 402(10419):2346–2394, December 2023. ISSN 0140-6736, 1474-547X. doi: 10.1016/S0140-6736(23)01859-7.
- [15] Tim Repke, **Max Callaghan**, William F. Lamb, Sarah Lück, Finn Müller-Hansen, and Jan C. Minx. Attention to climate change only temporarily diverted by COVID-19. *Submitted to Environmental Research Communications*, 2023.
- [16] S M Smith, O Geden, G F Nemet, M J Gidden, W F Lamb, C Powis, R Bellamy, **M W Callaghan**, A Cowie, E Cox, S Fuss, T Gasser, G Grassi, J Greene, S Lück, A Mohan, F Müller-Hansen, G P Peters, Y Pratama, T Repke, K Riahi, F Schenuit, J Steinhauser, J Streffler, J M

Valenzuela, and J C Minx. The State of Carbon Dioxide Removal - 1st Edition. Technical report, The State of Carbon Dioxide Removal, 2023.

- [17] Finn Müller-Hansen, Tim Repke, Chad M. Baum, Elina Brutschin, **Max W. Callaghan**, Ramit Debnath, William F. Lamb, Sean Low, Sarah Lück, Cameron Roberts, Benjamin K. Sovacool, and Jan C. Minx. Attention, sentiments and emotions towards emerging climate technologies on Twitter. *Global Environmental Change*, 83:102765, December 2023. ISSN 0959-3780. doi: 10.1016/j.gloenvcha.2023.102765.
- [18] Shihui Zhang, Chi Zhang, Wenjia Cai, Yuqi Bai, **Max Callaghan**, Nan Chang, Bin Chen, Huiqi Chen, Liangliang Cheng, Hancheng Dai, Xin Dai, Weicheng Fan, Xiaoyi Fang, Tong Gao, Yang Geng, Dabo Guan, Yixin Hu, Junyi Hua, Cunrui Huang, Hong Huang, Jianbin Huang, Xiaomeng Huang, John S. Ji, Qiaolei Jiang, Xiaopeng Jiang, Gregor Kiesewetter, Tiantian Li, Lu Liang, Borong Lin, Hualiang Lin, Huan Liu, Qiyong Liu, Xiaobo Liu, Zhao Liu, Zhu Liu, Yufu Liu, Bo Lu, Chenxi Lu, Zhenyu Luo, Wei Ma, Zhifu Mi, Chao Ren, Marina Romanello, Jianxiang Shen, Jing Su, Yuze Sun, Xinlu Sun, Xu Tang, Maria Walawender, Can Wang, Qing Wang, Rui Wang, Laura Warnecke, Wangyu Wei, Sanmei Wen, Yang Xie, Hui Xiong, Bing Xu, Yu Yan, Xiu Yang, Fanghong Yao, Le Yu, Jiacan Yuan, Yiping Zeng, Jing Zhang, Lu Zhang, Rui Zhang, Shangchen Zhang, Shaohui Zhang, Mengzhen Zhao, Dashan Zheng, Hao Zhou, Jingbo Zhou, Ziqiao Zhou, Yong Luo, and Peng Gong. The 2023 China report of the Lancet Countdown on health and climate change: Taking stock for a thriving future. *The Lancet Public Health*, 8 (12):e978–e995, December 2023. ISSN 2468-2667. doi: 10.1016/S2468-2667(23)00245-1.
- [19] Kim R. van Daalen, Marina Romanello, Joacim Rocklöv, Jan C. Semenza, Cathryn Tonne, Anil Markandya, Niheer Dasandi, Slava Jankin, Hicham Achebak, Joan Ballester, Hannah Bechara, **Max W. Callaghan**, Jonathan Chambers, Shouro Dasgupta, Paul Drummond, Zia Farooq, Olga Gasparyan, Nube Gonzalez-Reviriego, Ian Hamilton, Risto Hänninen, Aleksandra Kazmierczak, Vladimir Kendrovski, Harry Kennard, Gregor Kiesewetter, Simon J. Lloyd, Martin Lotto Batista, Jaime Martinez-Urtaza, Carles Milà, Jan C. Minx, Mark Nieuwenhuijsen, Julia Palamarchuk, Marcos Quijal-Zamorano, Elizabeth J. Z. Robinson, Daniel Scamman, Oliver Schmoll, Maquins Odhiambo Sewe, Henrik Sjödin, Mikhail Sofiev, Balakrishnan Salaraju-Murali, Marco Springmann, Joaquin Triñanes, Josep M. Anto, Maria Nilsson, and Rachel Lowe. The 2022 Europe report of the Lancet Countdown on health and climate change: Towards a climate resilient future. *The Lancet Public Health*, 7(11):e942–e965, November 2022. ISSN 2468-2667. doi: 10.1016/S2468-2667(22)00197-9.
- [20] Finn Müller-Hansen, Yuan Ting Lee, **Max Callaghan**, Slava Jankin, and Jan C. Minx. The German coal debate on Twitter: Reactions to a corporate policy process. *Energy Policy*, 169: 113178, October 2022. ISSN 0301-4215. doi: 10.1016/j.enpol.2022.113178.
- [21] Felix Creutzig, Leila Niamir, Xuemei Bai, **Max Callaghan**, Jonathan Cullen, Julio Díaz-José, Maria Figueroa, Arnulf Grubler, William F. Lamb, Adrian Leip, Eric Masanet, Érika Mata, Linus Mattauch, Jan C. Minx, Sebastian Mirasgedis, Yacob Mulugetta, Sudarmanto Budi Nugroho, Minal Pathak, Patricia Perkins, Joyashree Roy, Stephane de la Rue du Can, Yamina Saheb, Shreya Some, Linda Steg, Julia Steinberger, and Diana Ürge-Vorsatz. Demand-side solutions to climate change mitigation consistent with high levels of well-being. *Nature Climate Change*, 12 (1):36–46, January 2022. ISSN 1758-6798. doi: 10.1038/s41558-021-01219-y.

- [22] **Max Callaghan**, Shraddha Vasudevan, Lucy Banisch, Niklas Doebbeling, Duncan Edmondson, Christian Flachsland, William Lamb, Sebastian Levi, Finn Müller-Hansen, Eduardo Posada, and Jan Christoph Minx. What is the evidence on climate mitigation policies, and to what extent can it be identified and classified using Machine Learning? A machine-learning-assisted systematic map protocol. 2022. doi: 10.21203/rs.3.pex-1910/v1.
- [23] Marina Romanello, Claudia Di Napoli, Paul Drummond, Carole Green, Harry Kennard, Pete Lampard, Daniel Scamman, Nigel Arnell, Sonja Ayeb-Karlsson, Lea Berrang Ford, Kristine Belesova, Kathryn Bowen, Wenjia Cai, **Max Callaghan**, Diarmid Campbell-Lendrum, Jonathan Chambers, Kim R. van Daalen, Carole Dalin, Niheer Dasandi, Shouro Dasgupta, Michael Davies, Paula Dominguez-Salas, Robert Dubrow, Kristie L. Ebi, Matthew Eckelman, Paul Ekins, Luis E. Escobar, Lucien Georgeson, Hilary Graham, Samuel H. Gunther, Ian Hamilton, Yun Hang, Risto Hänninen, Stella Hartinger, Kehan He, Jeremy J. Hess, Shih-Che Hsu, Slava Jankin, Louis Jamart, Ollie Jay, Ilan Kelman, Gregor Kiesewetter, Patrick Kinney, Tord Kjellstrom, Dominic Kniveton, Jason K. W. Lee, Bruno Lemke, Yang Liu, Zhao Liu, Melissa Lott, Martin Lotto Batista, Rachel Lowe, Frances MacGuire, Maquins Odhiambo Sewe, Jaime Martinez-Urtaza, Mark Maslin, Lucy McAllister, Alice McGushin, Celia McMichael, Zhifu Mi, James Milner, Kelton Minor, Jan C. Minx, Nahid Mohajeri, Maziar Moradi-Lakeh, Karyn Morrissey, Simon Munzert, Kris A. Murray, Tara Neville, Maria Nilsson, Nick Obradovich, Megan B. O'Hare, Tadj Oreszczyn, Matthias Otto, Fereidoon Owfi, Olivia Pearman, Mahnaz Rabbaniha, Elizabeth J. Z. Robinson, Joacim Rocklöv, Renee N. Salas, Jan C. Semenza, Jodi D. Sherman, Liuhua Shi, Joy Shumake-Guillemot, Grant Silbert, Mikhail Sofiev, Marco Springmann, Jennifer Stowell, Meisam Tabatabaei, Jonathon Taylor, Joaquin Triñanes, Fabian Wagner, Paul Wilkinson, Matthew Winning, Marisol Yglesias-González, Shihui Zhang, Peng Gong, Hugh Montgomery, and Anthony Costello. The 2022 report of the Lancet Countdown on health and climate change: Health at the mercy of fossil fuels. *The Lancet*, 400(10363):1619–1654, November 2022. ISSN 0140-6736, 1474-547X. doi: 10.1016/S0140-6736(22)01540-9.
- [24] Wenjia Cai, Chi Zhang, Shihui Zhang, Yuqi Bai, **Max Callaghan**, Nan Chang, Bin Chen, Huiqi Chen, Liangliang Cheng, Xueqin Cui, Hancheng Dai, Bawuerjiang Danna, Wenxuan Dong, Weicheng Fan, Xiaoyi Fang, Tong Gao, Yang Geng, Dabo Guan, Yixin Hu, Junyi Hua, Cunrui Huang, Hong Huang, Jianbin Huang, Linlang Jiang, Qiaolei Jiang, Xiaopeng Jiang, Hu Jin, Gregor Kiesewetter, Lu Liang, Borong Lin, Hualiang Lin, Huan Liu, Qiyong Liu, Tao Liu, Xiaobo Liu, Xinyuan Liu, Zhao Liu, Zhu Liu, Shuhan Lou, Chenxi Lu, Zhenyu Luo, Wenjun Meng, Hui Miao, Chao Ren, Marina Romanello, Wolfgang Schöpp, Jing Su, Xu Tang, Can Wang, Qiong Wang, Laura Warnecke, Sanmei Wen, Wilfried Winiwarter, Yang Xie, Bing Xu, Yu Yan, Xiu Yang, Fanghong Yao, Le Yu, Jiacan Yuan, Yiping Zeng, Jing Zhang, Lu Zhang, Rui Zhang, Shangchen Zhang, Shaohui Zhang, Qi Zhao, Dashan Zheng, Hao Zhou, Jingbo Zhou, Margaret Fu-Chun Chan Fung, Yong Luo, and Peng Gong. The 2022 China report of the Lancet Countdown on health and climate change: Leveraging climate actions for healthy ageing. *The Lancet Public Health*, 0(0), October 2022. ISSN 2468-2667. doi: 10.1016/S2468-2667(22)00224-9.
- [25] Felix Creutzig, **Max Callaghan**, Anjali Ramakrishnan, Aneeqe Javaid, Leila Niamir, Jan Minx, Finn Müller-Hansen, Benjamin Sovacool, Zakia Afroz, Mark Andor, Miklos Antal, Victor Court, Nandini Das, Julio Díaz-José, Friederike Döbbe, Maria J. Figueroa, Andrew Gouldson, Helmut Haberl, Andrew Hook, Diana Ivanova, William F. Lamb, Nadia Maïzi, Érika Mata, Kristian S. Nielsen, Chioma Daisy Onyige, Lucia A. Reisch, Joyashree Roy, Pauline Scheelbeek, Mahendra Sethi, Shreya Some, Steven Sorrell, Mathilde Tessier, Tania Urmee, Doris Virág, Can Wan,

Dominik Wiedenhofer, and Charlie Wilson. Reviewing the scope and thematic focus of 100,000 publications on energy consumption, services and social aspects of climate change: A big data approach to demand-side mitigation. *Environmental Research Letters*, 16(3):033001, February 2021. ISSN 1748-9326. doi: 10.1088/1748-9326/abd78b.

- [26] **Max Callaghan**, Carl-Friedrich Schleussner, Shruti Nath, Quentin Lejeune, Thomas R. Knutson, Markus Reichstein, Gerrit Hansen, Emily Theokritoff, Marina Andrijevic, Robert J. Brecha, Michael Hegarty, Chelsea Jones, Kaylin Lee, Agathe Lucas, Nicole van Maanen, Inga Menke, Peter Pfleiderer, Burcu Yesil, and Jan C. Minx. Machine-learning-based evidence and attribution mapping of 100,000 climate impact studies. *Nature Climate Change*, 11(11):966–972, November 2021. ISSN 1758-6798. doi: 10.1038/s41558-021-01168-6.
- [27] Francesca Diluio, Paula Walk, Niccolò Manych, Nicola Cerutti, Vladislav Chipiga, Annabelle Workman, Ceren Ayas, Ryna Yiyun Cui, Diyang Cui, Kaihui Song, Lucy A. Banisch, Nikolaj Moretti, **Max W. Callaghan**, Leon Clarke, Felix Creutzig, Jérôme Hilaire, Frank Jotzo, Matthias Kalkuhl, William F. Lamb, Andreas Löschel, Finn Müller-Hansen, Gregory F. Nemet, Pao-Yu Oei, Benjamin K. Sovacool, Jan C. Steckel, Sebastian Thomas, John Wiseman, and Jan C. Minx. Coal transitions—part 1: A systematic map and review of case study learnings from regional, national, and local coal phase-out experiences. *Environmental Research Letters*, 16(11):113003, October 2021. ISSN 1748-9326. doi: 10.1088/1748-9326/ac1b58.
- [28] Lea Berrang-Ford, Anne J. Sietsma, **Max Callaghan**, Jan C. Minx, Pauline F. D. Scheelbeek, Neal R. Haddaway, Andy Haines, and Alan D. Dangour. Systematic mapping of global research on climate and health: A machine learning review. *The Lancet Planetary Health*, 5(8):e514–e525, August 2021. ISSN 2542-5196. doi: 10.1016/S2542-5196(21)00179-0.
- [29] Vivien Fisch-Romito, Celine Guivarch, Felix Creutzig, Jan C. Minx, and **Max W. Callaghan**. Systematic map of the literature on carbon lock-in induced by long-lived capital. *Environmental Research Letters*, April 2021. ISSN 1748-9326. doi: 10.1088/1748-9326/aba660.
- [30] Tarun M. Khanna, Giovanni Baiocchi, **Max Callaghan**, Felix Creutzig, Horia Guias, Neal R. Haddaway, Lion Hirth, Aneeqe Javaid, Nicolas Koch, Sonja Laukemper, Andreas Löschel, Maria del Mar Zamora Dominguez, and Jan C. Minx. A multi-country meta-analysis on the role of behavioural change in reducing energy consumption and CO2 emissions in residential buildings. *Nature Energy*, 6(9):925–932, September 2021. ISSN 2058-7546. doi: 10.1038/s41560-021-00866-x.
- [31] Lea Berrang-Ford, A. R. Siders, Alexandra Lesnikowski, Alexandra Paige Fischer, **Max W. Callaghan**, Neal R. Haddaway, Katharine J. Mach, Malcolm Araos, Mohammad Aminur Rahman Shah, Mia Wannewitz, Deepal Doshi, Timo Leiter, Custodio Matavel, Justice Issah Musah-Surugu, Gabrielle Wong-Parodi, Philip Antwi-Agyei, Idowu Ajibade, Neha Chauhan, William Kakenmaster, Caitlin Grady, Vasiliki I. Chalastani, Kripa Jagannathan, Eranga K. Galappaththi, Asha Sitati, Giulia Scarpa, Edmond Totin, Katy Davis, Nikita Charles Hamilton, Christine J. Kirchhoff, Praveen Kumar, Brian Pentz, Nicholas P. Simpson, Emily Theokritoff, Delphine Deryng, Diana Reckien, Carol Zavaleta-Cortijo, Nicola Ulibarri, Alcade C. Segnon, Vhalinavho Khavhagali, Yuanyuan Shang, Luckson Zvobgo, Zinta Zommers, Jiren Xu, Portia Adade Williams, Ivan Villaverde Canosa, Nicole van Maanen, Bianca van Bavel, Maarten van Aalst, Lynée L. Turek-Hankins, Hasti Trivedi, Christopher H. Trisos, Adelle Thomas, Shinny Thakur, Sienna Templeman, Lindsay C. Stringer, Garry Sotnik, Kathryn Dana Sjostrom, Chandni

Singh, Mariella Z. Siña, Roopam Shukla, Jordi Sardans, Eunice A. Salubi, Lolita Shaila Safaee Chalkasra, Raquel Ruiz-Díaz, Carys Richards, Pratik Pokharel, Jan Petzold, Josep Penuelas, Julia Pelaez Avila, Julia B. Pazmino Murillo, Souha Ouni, Jennifer Niemann, Miriam Nielsen, Mark New, Patricia Nayna Schwerdtle, Gabriela Nagle Alverio, Cristina A. Mullin, Joshua Mullenite, Anuska Mosurska, Mike D. Morecroft, Jan C. Minx, Gina Maskell, Abraham Marshall Nunbogu, Alexandre K. Magnan, Shuaib Lwasa, Megan Lukas-Sithole, Tabea Lissner, Oliver Lilford, Steven F. Koller, Matthew Jurjonas, Elphin Tom Joe, Lam T. M. Huynh, Avery Hill, Rebecca R. Hernandez, Greeshma Hegde, Tom Hawxwell, Sherilee Harper, Alexandra Harden, Marjolijn Haasnoot, Elisabeth A. Gilmore, Leah Gichuki, Alyssa Gatt, Matthias Garschagen, James D. Ford, Andrew Forbes, Aidan D. Farrell, Carolyn A. F. Enquist, Susan Elliott, Emily Duncan, Erin Coughlan de Perez, Shaugn Coggins, Tara Chen, Donovan Campbell, Katherine E. Browne, Kathryn J. Bowen, Robbert Biesbroek, Indra D. Bhatt, Rachel Bezner Kerr, Stephanie L. Barr, Emily Baker, Stephanie E. Austin, Ingrid Arotoma-Rojas, Christa Anderson, Warda Ajaz, Tanvi Agrawal, and Thelma Zulfawu Abu. A systematic global stocktake of evidence on human adaptation to climate change. *Nature Climate Change*, 11(11):989–1000, November 2021. ISSN 1758-6798. doi: 10.1038/s41558-021-01170-y.

- [32] Anne J. Sietsma, James D. Ford, **Max W. Callaghan**, and Jan C. Minx. Progress in climate change adaptation research. *Environmental Research Letters*, 16(5):054038, April 2021. ISSN 1748-9326. doi: 10.1088/1748-9326/abf7f3.
- [33] Pauline F. D. Scheelbeek, Alan D. Dangour, Stephanie Jarmul, Grace Turner, Anne J. Sietsma, Jan C. Minx, **Max Callaghan**, Idowu Ajibade, Stephanie E. Austin, Robbert Biesbroek, Kathryn J. Bowen, Tara Chen, Katy Davis, Tim Ensor, James D. Ford, Eranga K. Galappaththi, Elphin T. Joe, Issah J. Musah-Surugu, Gabriela Nagle Alverio, Patricia Nayna Schwerdtle, Pratik Pokharel, Eunice A. Salubi, Giulia Scarpa, Alcade C. Segnon, Mariella Siña, Sienna Templeman, Jiren Xu, Carol Zavaleta-Cortijo, and Lea Berrang-Ford. The effects on public health of climate change adaptation responses: A systematic review of evidence from low- and middle-income countries. *Environmental Research Letters*, 16(7):073001, July 2021. ISSN 1748-9326. doi: 10.1088/1748-9326/ac092c.
- [34] Finn Müller-Hansen, **Max W. Callaghan**, Yuan Ting Lee, Anna Leipprand, Christian Flachsland, and Jan C. Minx. Who cares about coal? Analyzing 70 years of German parliamentary debates on coal with dynamic topic modeling. *Energy Research & Social Science*, 72:101869, February 2021. ISSN 2214-6296. doi: 10.1016/j.erss.2020.101869.
- [35] Neal R. Haddaway, **Max W. Callaghan**, Alexandra M. Collins, William F. Lamb, Jan C. Minx, James Thomas, and Denny John. On the use of computer-assistance to facilitate systematic mapping. *Campbell Systematic Reviews*, 16(4):e1129, November 2020. ISSN 1891-1803. doi: 10.1002/cl2.1129.
- [36] Finn Müller-Hansen, **Max W. Callaghan**, and Jan C. Minx. Text as big data: Develop codes of practice for rigorous computational text analysis in energy social science. *Energy Research & Social Science*, 70:101691, December 2020. ISSN 2214-6296. doi: 10.1016/j.erss.2020.101691.
- [37] **Max Callaghan** and Finn Müller-Hansen. Statistical Stopping Criteria for Automated Screening in Systematic Reviews. *Systematic Reviews*, September 2020. doi: 10.21203/rs.2.18218/v2.

- [38] **Max W. Callaghan**, Jan C. Minx, and Piers M. Forster. A topography of climate change research. *Nature Climate Change*, 10(2):118–123, February 2020. ISSN 1758-6798. doi: 10.1038/s41558-019-0684-5.
- [39] Diana Ivanova, John Barrett, Dominik Wiedenhofer, Biljana Macura, **Max Callaghan**, and Felix Creutzig. Quantifying the potential for climate change mitigation of consumption options. *Environmental Research Letters*, 15(9):093001, August 2020. ISSN 1748-9326. doi: 10.1088/1748-9326/ab8589.
- [40] William F. Lamb, Felix Creutzig, **Max W. Callaghan**, and Jan C. Minx. Learning about urban climate solutions from case studies. *Nature Climate Change*, 9(4):279–287, April 2019. ISSN 1758-6798. doi: 10.1038/s41558-019-0440-x.
- [41] Jérôme Hilaire, Jan C. Minx, **Max W. Callaghan**, Jae Edmonds, Gunnar Luderer, Gregory F. Nemet, Joeri Rogelj, and Maria del Mar Zamora. Negative emissions and international climate goals—learning from and about mitigation scenarios. *Climatic Change*, 157(2):189–219, November 2019. ISSN 1573-1480. doi: 10.1007/s10584-019-02516-4.
- [42] Jan C. Minx, William F. Lamb, **Max W. Callaghan**, Sabine Fuss, Jérôme Hilaire, Felix Creutzig, Thorben Amann, Tim Beringer, Wagner de Oliveira Garcia, Jens Hartmann, Tarun Khanna, Dominic Lenzi, Gunnar Luderer, Gregory F. Nemet, Joeri Rogelj, Pete Smith, Jose Luis Vicente Vicente, Jennifer Wilcox, and Maria del Mar Zamora Dominguez. Negative emissions—Part 1: Research landscape and synthesis. *Environmental Research Letters*, 13(6):063001, May 2018. ISSN 1748-9326. doi: 10.1088/1748-9326/aabf9b.
- [43] William F. Lamb, **Max W. Callaghan**, Felix Creutzig, Radhika Khosla, and Jan C. Minx. The literature landscape on 1.5[deg]C climate change and cities. *Current Opinion in Environmental Sustainability*, 30:26–34, February 2018. ISSN 1877-3435. doi: 10.1016/j.cosust.2018.02.008.
- [44] Gregory F. Nemet, **Max W. Callaghan**, Felix Creutzig, Sabine Fuss, Jens Hartmann, Jérôme Hilaire, William F. Lamb, Jan C. Minx, Sophia Rogers, and Pete Smith. Negative emissions—Part 3: Innovation and upscaling. *Environmental Research Letters*, 13(6):063003, May 2018. ISSN 1748-9326. doi: 10.1088/1748-9326/aabff4.
- [45] Sabine Fuss, William F. Lamb, **Max W. Callaghan**, Jérôme Hilaire, Felix Creutzig, Thorben Amann, Tim Beringer, Wagner de Oliveira Garcia, Jens Hartmann, Tarun Khanna, Gunnar Luderer, Gregory F. Nemet, Joeri Rogelj, Pete Smith, José Luis Vicente Vicente, Jennifer Wilcox, Maria del Mar Zamora Dominguez, and Jan C. Minx. Negative emissions—Part 2: Costs, potentials and side effects. *Environmental Research Letters*, 13(6):063002, May 2018. ISSN 1748-9326. doi: 10.1088/1748-9326/aabf9f.
- [46] Jan C. Minx, **Max Callaghan**, William F. Lamb, Jennifer Garard, and Ottmar Edenhofer. Learning about climate change solutions in the IPCC and beyond. *Environmental Science & Policy*, 77:252–259, November 2017. ISSN 1462-9011. doi: 10.1016/j.envsci.2017.05.014.
- [47] Jan C. Minx, William F. Lamb, **Max W. Callaghan**, Lutz Bornmann, and Sabine Fuss. Fast growing research on negative emissions. *Environmental Research Letters*, 12(3):035007, March 2017. ISSN 1748-9326. doi: 10.1088/1748-9326/aa5ee5.
- [48] Christoph von Stechow, Jan C. Minx, Keywan Riahi, Jessica Jewell, David L. McCollum, **Max W. Callaghan**, Christoph Bertram, Gunnar Luderer, and Giovanni Baiocchi. 2[deg]C and SDGs:

United they stand, divided they fall? *Environmental Research Letters*, 11(3):034022, March 2016. ISSN 1748-9326. doi: 10.1088/1748-9326/11/3/034022.

Non peer reviewed publications

- July 2022 **Here are the most effective things you can do to fight climate change**, *Guest post for the Conversation*, link
- November 2021 **What 100,000 studies tell us about climate impacts around the world**, *Guest post for Carbon Brief*, link

Software

- NACSOS **A Django site for managing collections of documents, screening or coding them, and doing NLP tasks with them like topic modelling or classification**, <https://doi.org/10.5281/zenodo.4121526>
- Climate and Health Map **An interactive site built with Dash to present the results of a machine learning-assisted systematic map of climate and health literature**, <https://apsis.mcc-berlin.net/climate-health/>
- The Homero-manteion **A digitized version of the ancient method of divination using Homeric poetry**, <http://homeromanteion.com/>
- BuscarR **An R package to support safe early stopping during machine learning prioritised screening**, <https://mcallaghan.github.io/buscarR/>
- Buscarpy **A Python package to support safe early stopping during machine learning prioritised screening**, <https://buscarpy.readthedocs.io/>

Conferences and Talks

- Poster Presentation **The safe use of LLMs for automated screening**, *Global Evidence Summit*, Prague, September 11 2024
- Invited Keynote **Opportunities and challenges in using NLP to assist evidence synthesis**, *Text-as-data in Hydrology, Natural Hazards, and Climate Change Research*, Potsdam, March 1 2024
- Poster Presentation **BUSCAR: Biased Urn-based Stopping Criteria for technology Assisted Reviews**, *Cochrane Colloquium*, London, September 6 2023
- Invited Talk **Natural Language Processing for climate change texts at the science-policy interface**, *Climate Change AI Berlin Brandenburg*, Berlin, July 6 2022
- Invited Webinar **Machine Reading the Growing Climate Science and Adaptation Literature**, *Climate Change AI*, Online, May 13 2022
- Invited Webinar **A database of 100,000 studies on attributable climate impacts derived with machine learning: methodology and potential uses or follow-up analyses**, *PROCLIAS - Process-based models for climate impact attribution across sectors*, Online, May 09 2022
- Invited Talk **Statistical stopping criterial for automated screening in systematic reviews**, *Evidence Synthesis & Meta-Analysis in R*, Online Conference, January 18 2021

Conference **Machine learning for systematically mapping the climate change literature**,
Presentation *Applied Machine Learning Days*, EPFL, Lausanne, February 01 2020

Conference **A Topography of Climate Change Research**, *EGU*, Vienna, April 09 2019
Presentation

Conference **A Topography of Climate Change Research**, *EASST*, Lancaster, July 26 2018
Presentation